EXP NO: 5 DATE:	**Cloud Object Storage Manager Using Python & MinIO**

Aim:

To develop a Python-based application that interacts with a local MinIO server (an S3-compatible cloud storage emulator) to understand object storage concepts such as buckets, unique object keys, and metadata tracking.

Problem Statement:

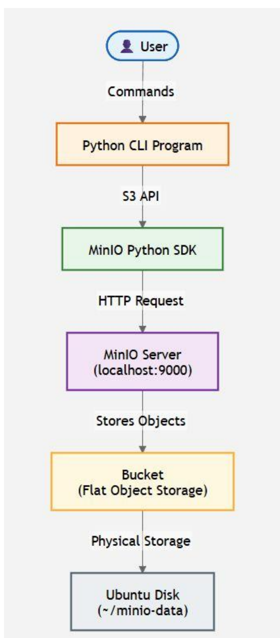
Develop a Python application that mimics cloud object storage behavior using an open-source MinIO server running locally inside a VirtualBox VM, communicating via the official MinIO Python API. The application must implement:

- Create Bucket – initialize a flat container on the MinIO server.
- Upload Object – upload a local file into a bucket with a unique key.
- Download Object – retrieve an object using its unique key.
- List Bucket Contents – display objects with metadata (size, last modified).
- Delete Object – remove an object using its unique key.

Pre-Requisites:

- Windows 10 Host Machine
- Oracle VM VirtualBox (Open Source)
- Ubuntu Desktop ISO image (Open Source)

Architecture:



Procedure:

Step 1-2: Update Ubuntu & Install Python

```
sudo apt upgrade -y
sudo apt install python3 python3-pip -y
python3 --version
pip3 –version
```

Step 3: Install MinIO Server

```
mkdir ~/minio && cd ~/minio
wget https://dl.min.io/server/minio/release/linux-amd64/minio
chmod +x minio
sudo mv minio /usr/local/bin/
minio –version
```

Step 4-5: Create Storage Folder & Start Server

```
mkdir ~/minio-data
export MINIO_ROOT_USER=minioadmin
export MINIO_ROOT_PASSWORD=minioadmin
minio server ~/minio-data --console-address ":9001"
```

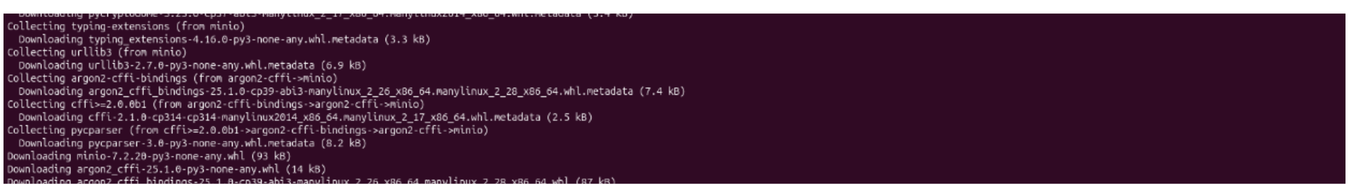
API: http://127.0.0.1:9000 | Console: http://127.0.0.1:9001 | Login: minioadmin / minioadmin

Step 6: Open MinIO Console

Visit http://localhost:9001 and log in. Initially there are no buckets.

Step 7-8: Create Project & Virtual Environment

```
mkdir ObjectStorageSimulator && cd ObjectStorageSimulator
python3 -m venv venv
source venv/bin/activate
```

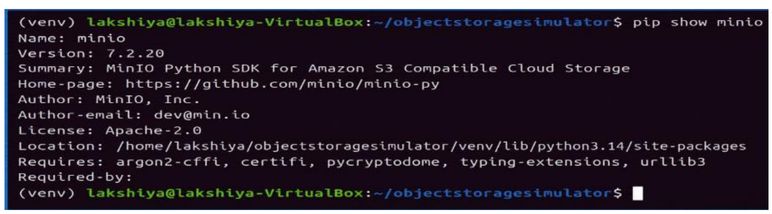


Step 9: Install MinIO Python SDK

```
pip install minio
```

Verify installation:

```
pip show minio
```

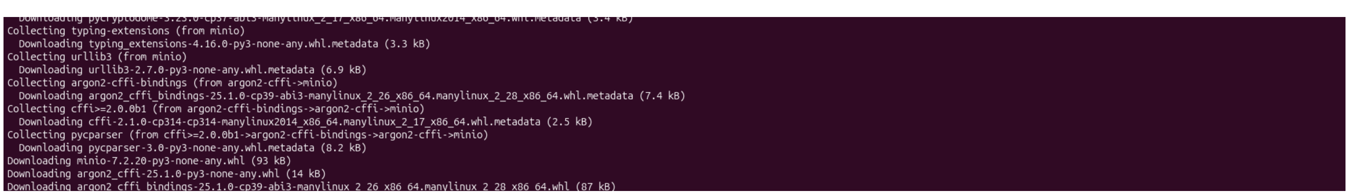


Step 10: Create Sample File & Bucket

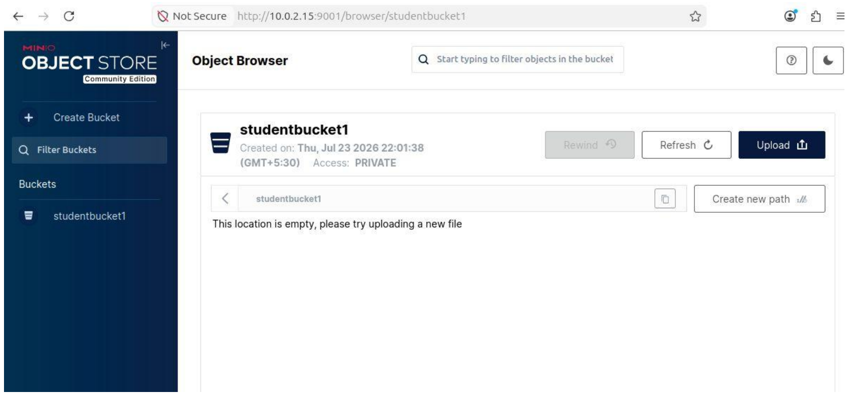
```
echo "Hello this is sample minio project" > sample.txt
```

Create main.py with the bucket-creation code below and run it:

```
from minio import Minio
client = Minio("localhost:9000", access_key="minioadmin",
    secret_key="minioadmin", secure=False)
bucket_name = "studentbucket"
if not client.bucket_exists(bucket_name):
    client.make_bucket(bucket_name)
    print("Bucket Created")
else:
    print("Bucket Already Exists")
```

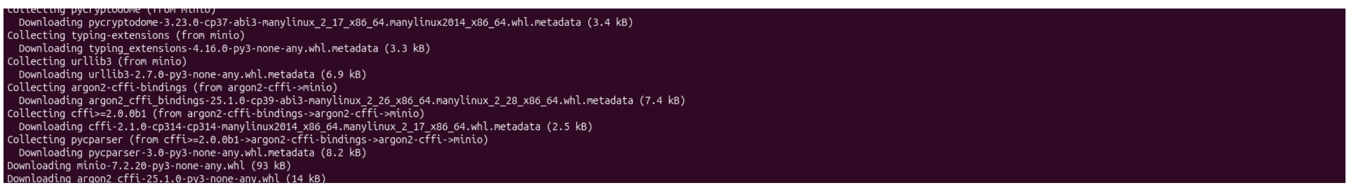


Verify in the MinIO console — the new bucket appears with no objects yet:

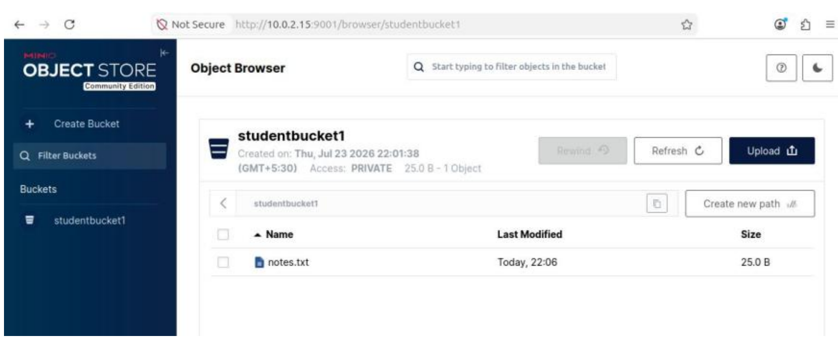


Step 11: Upload an Object

```
client.fput_object("studentbucket", "notes.txt", "sample.txt")
print("Upload Successful")
```



Verify in the console — notes.txt now appears inside the bucket:



Step 12: Download an Object

Modify main.py to replace the upload code with:

Replace code.

Python

```python
from minio import Minio

client = Minio(
    "localhost:9000",
    access_key="minioadmin",
    secret_key="minioadmin",
    secure=False
)

client.fget_object(
    "studentbucket1",
    "notes.txt",
    "downloaded.txt"
)

print("Download Successful")
```

Result:

Thus, a Python application was developed to perform bucket creation, object upload, and object download using the MinIO S3-compatible local object storage server.